

Name: _____

Instructor: _____

CHE 225 – Introduction to Chemical Engineering Analysis

Test #1. Closed book – closed notes. Write all your answers in the provided paper. *Begin each numbered problem on a new page, show all your work, and box your final numerical answers.* Only write your answers on the front of each page and start each problem on a fresh page.

HONOR PLEDGE: "I have neither given nor received unauthorized aid on this test. I will not discuss the exam with anyone until after 5 pm on Friday, February 8, 20xx."

Your name (printed)

Your signature

Problem 1: _____ / 23

Problem 2: _____ / 27

Problem 3: _____ / 25

Problem 4: _____ / 25

Total: _____ / 100

Problem 1 (23 points)

A fuel gas containing 80 mole% of methane (CH₄) and the balance of ethane (C₂H₆) is burned completely using 50% excess dry air (~79 mole% N₂, ~21 mole% O₂). **Assume complete combustion.**

- (5 points) Draw and label the complete flow chart.
- (5 points) Perform the degree of freedom analysis using either **extent of reaction (EOR)** or **atomic balances**.
- (5 points) Given your result in part b) do you have enough information to calculate the molar composition of the product gas? If yes, state why. If not, what information you need or can assume on your own?
- (8 points) Write down all balance equations you need to calculate the molar composition of the product gas. **Do not solve.**

Problem 2 (27 points)

Linearization is a useful strategy to fit nonlinear dependencies to a line. For each of the functions listed below,

- How would you plot the data to obtain a straight line?
- What is the slope and the intercept (both expressed in terms of the constants)?

The variables on the left- and right-hand sides in the equations below represent the dependent and independent variables, respectively. The equations also contain some constants (those are noted in the assignment). The independent variable is to be plotted on the x-axis (*i.e.*, abscissa) while the dependent variable will be displayed on the y-axis (*i.e.*, ordinate). **Equation a) is used as an example for the answer format.** Complete your response for b) to g).

a) $y = ax^2 + b.$

Answer: Plot y versus x^2 ; slope = a, intercept = b.

b) (4.5 points) $y = a\sqrt{x} + b$ (a, b...constants)

c) (4.5 points) $q = a \exp(b \cdot g)$ (a, b...constants)

d) (4.5 points) $w = (a \cdot t^2 - b)^{1/3}$ (a, b...constants)

e) (4.5 points) $\sin(y) = x \cdot (a \cdot x + b)^{-2}$ (a, b...constants)

f) (4.5 points) Boltzmann distribution, $p = A e^{-\frac{E}{kT}}$, where k is the Boltzmann's constant, and A and E are constants.

g) (4.5 points) Power-law fluid, $\Gamma = k \dot{\gamma}^n$, where k and n are constants.

Problem 3 (25 points)

Least squares regression (without linearization) is to be used to estimate parameters K and a in equation (1),

$$S = \frac{a[L]_{\text{initial}}}{K + [L]_{\text{initial}}} \quad (1)$$

Use the experimental data shown in **Figure 1** as a reference. Describe the steps you will carry out to obtain estimates of the parameters K and a in cells B15 and B16, respectively, using Solver in Excel.

- (10 points) Two new columns (column C and D) are needed in order to complete the least square process, as shown in **Figure 1 (black box)**. What are the variables you will need to calculate in column C and D? What are the specific formulas you will enter in cell C2 and D2, respectively? Use cell ID (e.g., C2) instead of actual number (e.g., 17) for your formulas. For instance: C2=D2+3*\$B\$15 (note: this is only an example how to write the answer).
- (5 points) If you want to calculate your final least square in the Objective cell (D15), what would be the formula you will enter in D15?
- (10 points) Now you have opened the Solver user interface as shown in **Figure 1**. What would you enter in the “Set Objective” box (indicate cell ID)? What will be your computational goal, max, min, or value of __? What would you enter in the “By Changing Variable Cells” box (indicate cell ID)? Finally, which button would you hit to get the results?

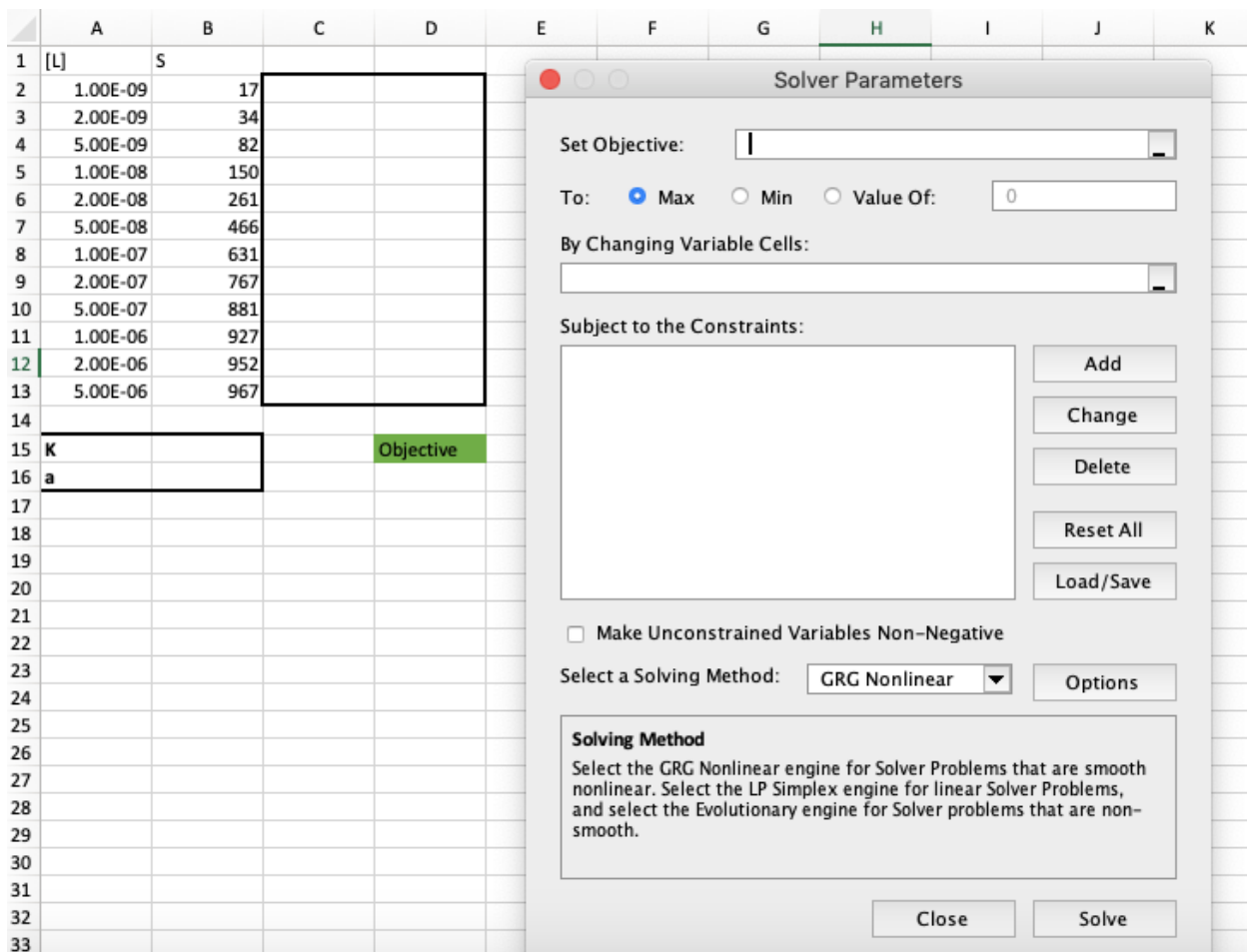


Figure 1: Finding K and a using Solver.

Problem 4 (25 points)

Consider the van der Waals equation of state:

$$\left[p + a \left(\frac{1}{\hat{V}} \right)^2 \right] (\hat{V} - b) = RT \quad (2)$$

where p is pressure (atm), T is absolute temperature (K), $\hat{V}$ is molecular volume, V/n (L/mol), R is the universal gas constant (L*atm/(mol*K)), and a , b are system-dependent constants.

- (5 points) What are the units of a and b in the above expression?
- (10 points) What is the expression of $p = f_{\text{xn}}(T)$, assuming that $\hat{V}$ is constant? What would you plot on the ordinate (*i.e.*, y-axis) and abscissa (*i.e.*, x-axis), in order to linearize the expression to determine a and b ? What is the slope? What is the intercept?
- (10 points) Assume that you are given T in °F and want to calculate p in Pa. How would you modify the expression of $p = f_{\text{xn}}(T)$, if the units of $\hat{V}$, R , a , and b remain the stated before?

Unit conversions & constants

Quantity	Conversion
Energy	1 J = 1 Nm = 10 ⁷ ergs = 10 ⁷ dyne·cm = 2.778 x 10 ⁻⁷ kWh
Force	1 N = 1 kg·m/s ² = 10 ⁵ dyne (or g·m/s ²)
Length	1 m = 100 cm = 39.37 in = 3.2808 ft
Mass	1 kg = 1000 g = 2.20462 lb _m
Power	1 W = 1 J/s = 0.23901 cal/s
Pressure	1 atm = 1.01325 x 10 ⁵ Pa (or N/m ²) = 760 mm Hg = 14.696 lb _f /in ² (or psi) = 1.01325 bars
Temperature	T(K) = T(°C)+273.15; T(°F) = 1.8·T(°C)+32
Volume	1 m ³ = 1000 liters = 10 ⁶ cm ³ = 35.3145 ft ³
Time	1 h = 60 min = 3600 s
Universal gas constant	8.314 J/K/mol 8.206 m ³ ·atm/K/mol 1.987 kcal/K/mol